VISHAL P.

CS SE The Consortium of Selective Schools in Essex

8:25 - 9:45.

# TEST 2

# MATHEMATICS

Name: _____

Candidate Number: _____

Primary School: _____

Boy or Girl: _____

Date of Birth: _____

Today's Date: _____

Test Taken At: _____

## READ THE FOLLOWING CAREFULLY:

1. Do not open this booklet until you are told to do so.
2. You may work the questions out in your head, or by writing on the white area around the question.
3. Work as quickly and as carefully as you can.
4. Make any alterations to your answers clearly. You will not lose marks for crossing out.
5. You will have 35 minutes to do the test. If you find you cannot do a question, do not waste time on it but go on to the next one.
6. Once the test has begun, you should not ask about questions in the test.
7. The use of electronic calculators of any description (including calculator watches) is NOT permitted.

| NOT TO BE FILLED IN BY PUPIL | | |
|---|---|---|
| PAGE | SCORE | |
| | R | W |
| 1 (5) | | |
| 2 (4) | | |
| 3 (3) | | |
| 4 (4) | | |
| 5 (5) | | |
| 6 (6) | | |
| 7 (6) | | |
| 8 (4) | | |
| TOTAL (37) | | |
| RATIO OF MARKERS: | | |

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2005 ENTRY

You have thirty five minutes to complete this paper.
Do your working out in the spaces on the paper.

Please do not write in this space

1. Calculate: 1000 - 276

Space for working:

Answer: _____

2. Calculate: $\frac{2}{3} \times 27$

Answer: _____

3. Calculate: 20% of £150

Answer: £ _____

4. Calculate: 723 x 29

Answer: _____

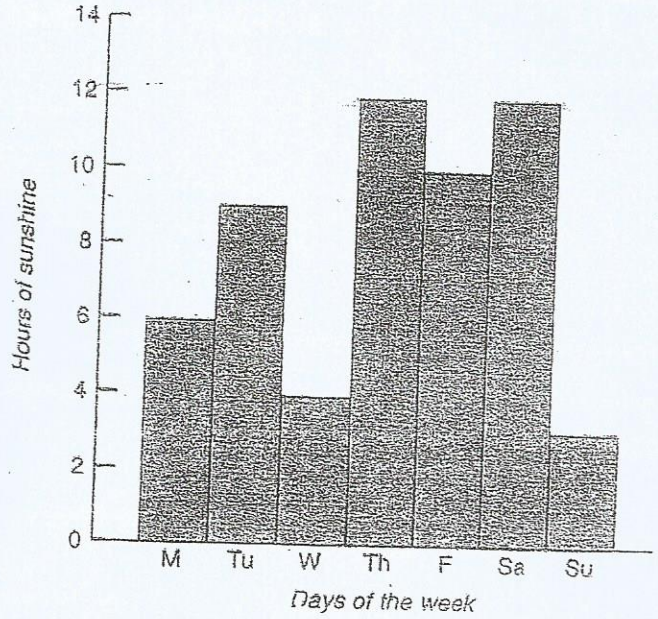
5. Which pair of numbers has the greatest difference?

A. (15 and 4)
B. (7 and -7)
C. (30 and 20)
D. (-5 and -14)
E. (10 and -3)

Answer: _____
(Choose A, B, C, D or E)

Please do not write in this space

6 The graph shows the number of hours of sunshine in Southend on each day of one week in the summer.




a) Which day was least sunny?

Answer: _____

b) Calculate the total number of hours of sun during this week.

Answer: _____ hours

c) What is the average number of hours of sunshine per day?

Answer: _____ hours

7 What is n if 6n - 5 = 37?

Answer: n = _____

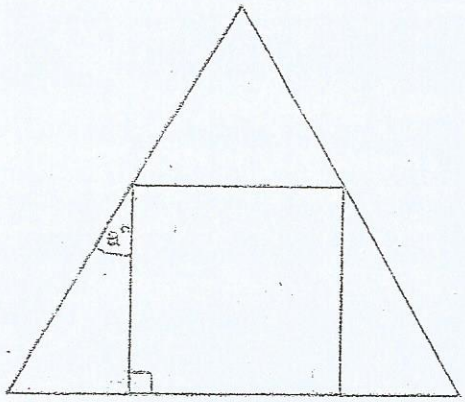
**8** This number sequence follows the rule "double, then subtract one".

0,95 0,9 0,8 0,6 ____ -0,6

What is the missing number?

Answer: _____

**9** The diagram shows a square drawn inside an equilateral triangle

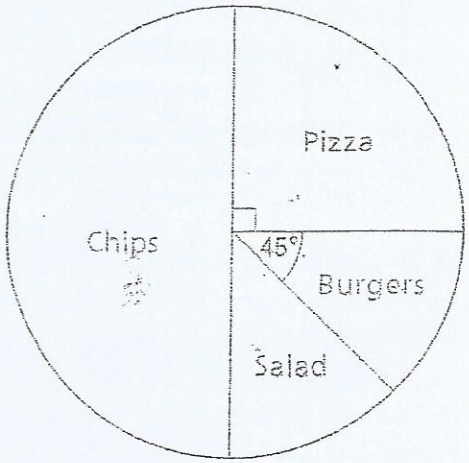



What is the size of the angle marked a°?

Answer: _____ °

**10** The piechart shows how 120 school children voted for their favourite item from the school menu.

How many children did NOT vote for salad?




Answer: _____ children

Please do not write in this space

**E2** Write the missing number in the box.

$78.39 = 70 + 8 + 0.3 +$ ☐

**E3** Calculate: $2\frac{3}{4} \div 1\frac{3}{8}$

Answer: _____

**E4** Calculate: $27.3 \div 7$

Answer: _____

**E5** Which of these is equal to $0.57 \times 10^2$ ?

A. 5.7
B. 570
C. $5.7 \times 10$
D. $0.057 \times 100$
E. $57 \div 10$

Answer: _____

(Choose A, B, C, D or E)

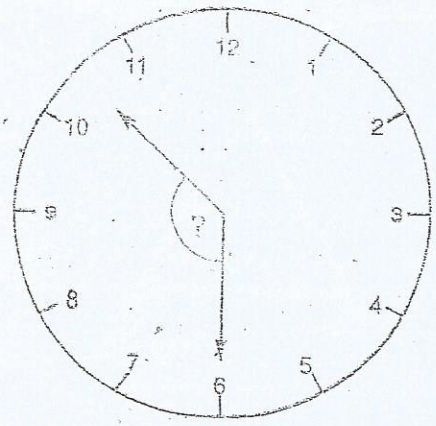
15 Which of these solid shapes has exactly five faces?

A. Cuboid
B. Cylinder
C. Tetrahedron
D. Rectangular Prism
E. Triangular Prism

Answer: _____

(Choose A, B, C, D or E)

16 What is the angle between the two hands of the clock at half past ten?




Answer: _____

17 I am 5 years older than my brother. Our ages add up to 19.
How old am I?

Answer: _____ years old

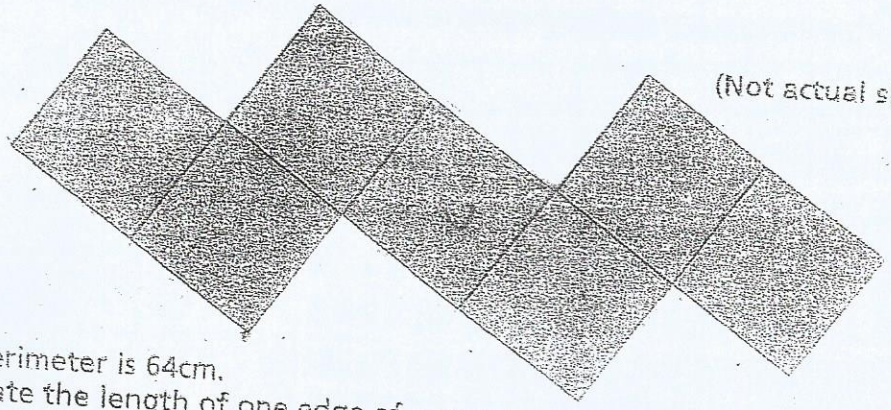
18 I have 20 large marbles.
12 of the large marbles are yellow.
I have 27 yellow marbles.
Altogether I have 44 large and small marbles.

Write the correct number of marbles in each of the 3 empty boxes.

| | *Large* | *Small* |
|---|---|---|
| *Not Yellow* | | |
| *Yellow* | 12 | |

PLEASE TURN OVER TO NEXT PAGE

19 This shape is made from 7 squares.

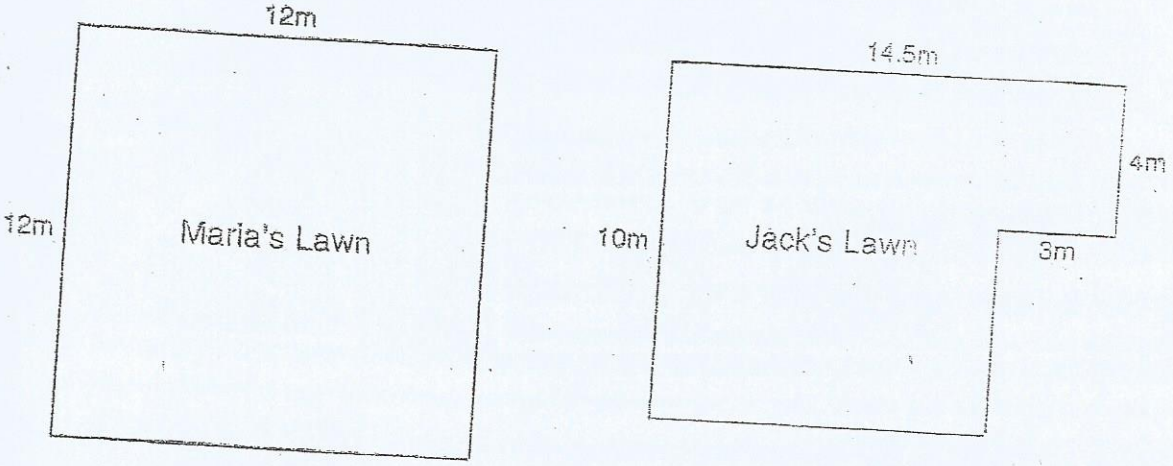


(Not actual size)

The perimeter is 64cm.
Calculate the length of one edge of a square.

Answer: _____ cm

20 How much larger is Maria's lawn than Jack's?




Area of Maria's Lawn =

Area of Jack's Lawn =

Answer: Difference in areas = _____ $m^2$

21 A full jar of home-made jam weighs 748g.
When half the jam has been eaten the weight is 480g.

How much will the empty jar weigh?

Answer: _____ g

22 The chart below shows three ways in which 5p, 2p and 1p coins can be used to make a total of 9p.

Add all the other ways that you can think of into the table.
How many different ways are there?

| 5p | 2p | 1p | Total Value |
|---|---|---|---|
| 1 | 2 | 0 | 9p |
| 0 | 0 | 9 | 9p |
| 0 | 4 | 1 | 9p |
| | | | |

Answer: Number of ways = _____

23 Look at the times of these two trains:

| | Train A | Train B |
|---|---|---|
| Leaves London | 18:09 | 18:49 |
| Arrives Cardiff | 20:25 | 20:45 |

Work out how long both journeys take.
How many minutes less than Train A does Train B take?

Train A takes_____ hrs _____mins

Train B takes_____ hrs _____mins

Answer: _____ minutes less

24 How many dots will there be in Pattern Seven?

Pattern One



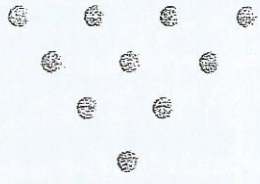
Pattern Two



Pattern Three



Pattern Four

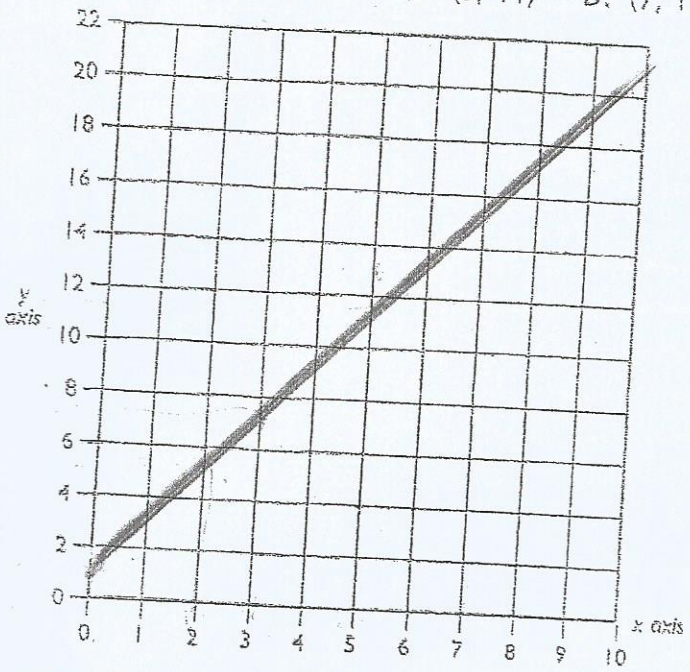


Answer: _____dots

24 Keira drew a straight line on a co-ordinate grid from the point (0, 1) to the point (10, 21). It will help for you to draw the line on the grid below.

Which of these points was <u>not</u> on Keira's line?

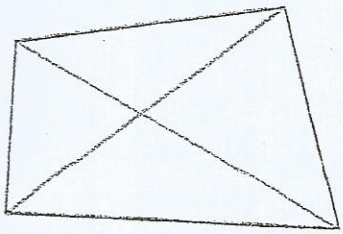
A. (2, 5) B. (3, 7) C. (5, 11) D. (7, 16) E. (8, 17)




Answer: _____

(Choose A, B, C, D or E)

25 A 4-sided polygon has a total of 2 diagonals as shown below.



If a polygon has a total of 9 diagonals
how many sides does it have?

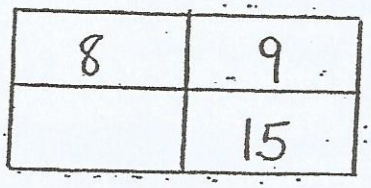
Answer: _____ sides

**END OF TEST – (You should have completed 26 questions.)**

Please do not write in this space

# MARKING SCHEME CSSE MATHS TEST

## FOR ENTRY IN SEPTEMBER 2005

| | | | |
|---|---|---|---|
| Page 1 | Q1 | 724 | 1 mark |
| | Q2 | 18 | 1 mark |
| | Q3 | £30 | 1 mark |
| | Q4 | 20967 | 1 mark |
| | Q5 | B | 1 mark |
| P2 | Q6 | a) Sunday<br>b) 56<br>c) 8 | 1 mark<br>1 mark<br>1 mark |
| | Q7 | n = 7 | 1 mark |
| P3 | Q8 | 0.2 | 1 mark |
| | Q9 | 30° | 1 mark |
| | Q10 | 105 | 1 mark |
| P4 | Q11 | 0.09 | 1 mark |
| | Q12 | 4 1/8 or 33/8 or 3 9/8 | 1 mark |
| | Q13 | 3.9 | 1 mark |
| | Q14 | C | 1 mark |
| P5 | Q15 | E | 1 mark |
| | Q16 | 135° or 225° | 1 mark |
| | Q17 | 12 | 1 mark |
| | Q18 |  <br>0 marks for 1 correct answer<br>1 mark for 2 correct answers<br>2 marks for 3 correct answers | 2 marks |
| P6 | Q19 | 4 | 2 marks |
| | Q20 | 17m²<br>Award 1 mark for correctly calculating the areas of Maria's and Jack's lawns as 144m² and 127m² | 2 marks |
| | Q21 | 212g | 2 marks |

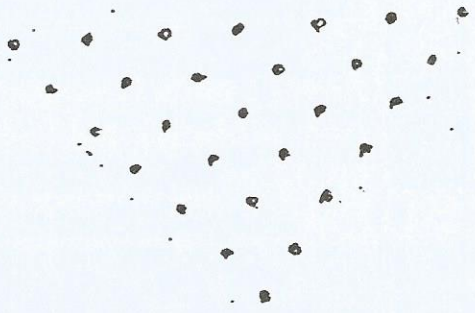
2 marks

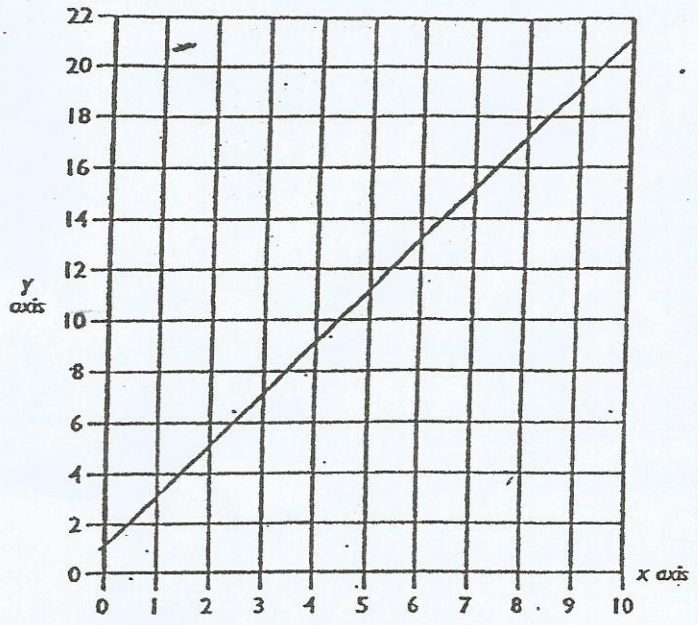
P7 Q22 Number of ways = 8

| 5p | 2p | 1p | Total Value |
|---|---|---|---|
| 1 | 2 | 0 | 9p |
| 0 | 0 | 9 | 9p |
| 0 | 4 | 1 | 9p |
| 1 | 1 | 2 | 9p |
| 1 | 0 | 4 | 9p |
| 0 | 3 | 3 | 9p |
| 0 | 2 | 5 | 9p |
| 0 | 1 | 7 | 9p |

The 5 lines added by candidates can be in any order.

Score no marks for the answer 8 with less than three correct lines added to the table.
Score 1 mark for adding 3, 4 or 5 correct lines to the table without a correct answer.
Score 2 marks for a correct table and a correct answer.

| | | | |
|---|---|---|---|
| | Q23 | 20 minutes<br>Train A takes 2 hrs 16 mins<br>Train B takes 1 hr 56 mins<br>Score 1 mark for correctly working out the times of the trains, or for giving the correct difference without the correct train times.<br>Score 2 marks for correctly working out the times of the trains and giving the correct difference. | 2 marks |
| | Q24 | 28 dots<br>Score 1 mark for correctly drawing pattern 7 only.<br>Score 2 marks for the answer 28, with or without the pattern.<br>  | 2 marks |

| | | | |
|---|---|---|---|
| P8 | Q25 | D<br>Award 1 mark for a correctly drawn line without a correct answer.<br>Award 2 marks for the answer D with or without the line being drawn.<br>  <br> | 2 marks |
| | Q26 | 6 sides | 2 marks |

Total marks 37